



ELIXIR User Guide

(Version 0.5)

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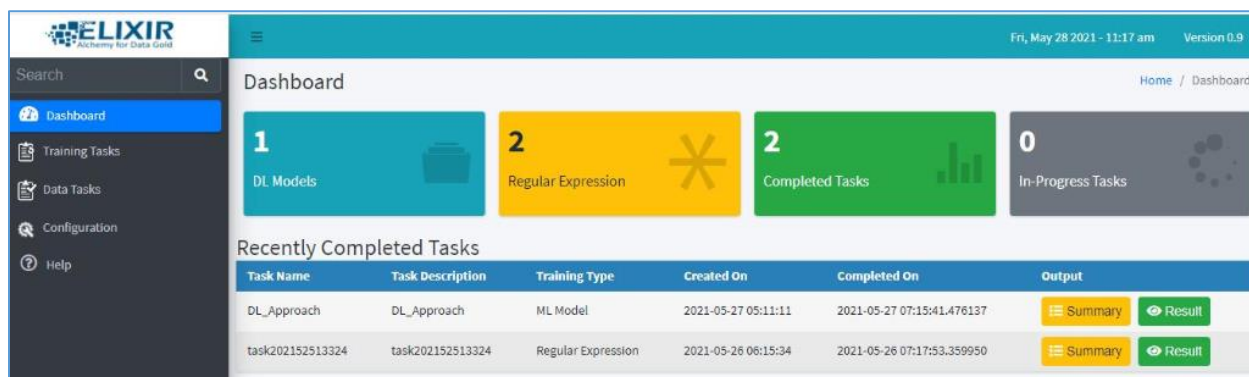
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Introduction

ELIXIR is a web application for executing training tasks to generate Regular Expression from log text and validate them. The tasks are specified on the datasets and are defined in Tags. We recommend Google Chrome browser for this application.

Dashboard

Dashboard is the landing page of the application and it displays the number of completed tasks, In-Progress Tasks, trained DL models, and regular expressions generated. You can also view the recently completed tasks, their summary and result.



Create Training Task

To create training task, follow the steps below:

1. In the left panel, click **Training Tasks**.
2. Training Tasks will be displayed as shown below.
3. Click **+Create Task**.

The first screenshot shows the 'Training Tasks' page. It features a table with columns: S.No., Task Name, Description, Result Type, Status, and Action. Two tasks are listed: 'DL_Approach' (ML Model, Completed) and 'task202152513324' (Regular Expression, Completed). Each task has buttons for Run, Edit, Summary, Result, and Delete. A '+ Create Task' button is in the top right.

The second screenshot shows the 'Create Training Task' form. It includes fields for Task Name (task202152865049), Description (Enter Training Task Description), and Log Separator (\n). There are dropdowns for Type and Import Type (Folder), and an Import Logs button.

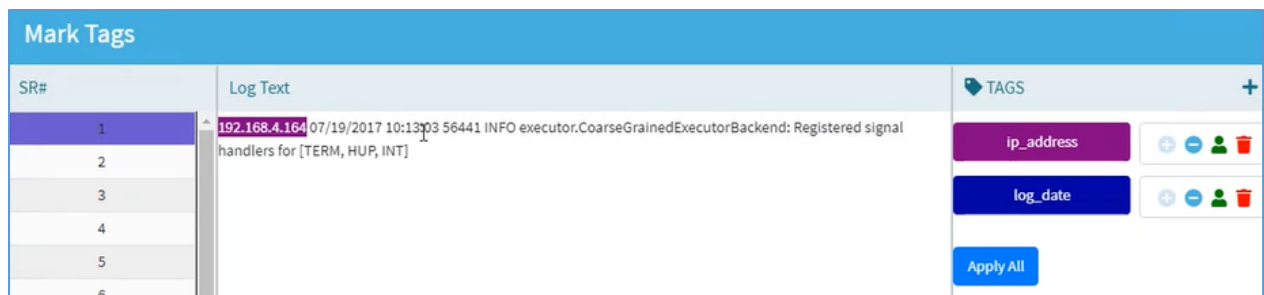
4. Enter the following fields:
 - a. **Task Name** - Enter a suitable task name.
 - b. **Description** - Enter a relevant task description.
 - c. **Log Separator** - Enter a log separator. For Example: \n, \$ etc.
 - d. **Type** - Select Regular Expression.
 - Based on the type selection, the result gets generated.
 - e. **Import Type** - Select Folder or File.

Note: If you enter a \$ log separator, make sure the \$ symbol is used in the log file.
 If you enter \n log separator, make sure there are a number of rows in the log file.
5. Click **Import** and select the log file.
6. As a result, you can see the lines (Log Text) of the log.

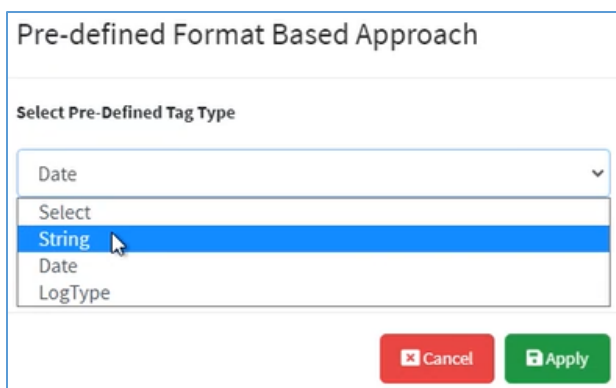
Define Tags

To define the tags, follow the steps below:

1. In the Create Training Task, click + icon on the right side of the log text.
2. The **Mark Tags** window opens up.



3. To add tags, click + icon on the right side of the TAGS.
4. After you add the tags, select a part of the log text and click the + icon on the right side of the tag created. For example: log_date.
5. As a result, the tags are defined and the corresponding parts of the log text are highlighted.
6. If required, click – icon on the right side of the available tag to delete the tag.
7. Similarly, click **User Defined Formats** icon and select any one of the options listed below.
 - a. Select **String**, enter the Starting Text and Ending Text.
 - b. Select **LogType**.
 - c. Select **Date** and select Format.
 - d. Click **Apply**.



Pre-defined Format Based Approach

Select Pre-Defined Tag Type

Date

Select

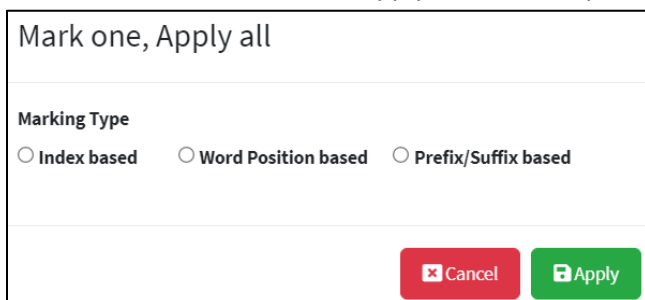
String

Date

LogType

Cancel Apply

8. If required, click **Delete** icon to delete the tag.
9. Click **Save**.
10. If required, click **Apply All** to apply the same tags for all the lines of log text.
11. As a result, The Mark one apply all window opens up.



Mark one, Apply all

Marking Type

☐ Index based ☐ Word Position based ☐ Prefix/Suffix based

Cancel Apply

12. If you select **Word Position Based**, the word position based approach screen appears.
 - a. Select **Single Word**.
 - b. Enter 1 if you want to mark the first word.
 - c. Similarly, you can enter any number to mark the word based on its position.
 - d. Click **Apply** to apply the single word tag for all the log text.
 - (OR)
 - e. Select **Multiple Words**.
 - f. Enter 1 in **Start Word** and 4 in **End Word** to mark first four words.
 - g. Similarly, you can mark any words continuously based on their position.
 - h. Click **Apply** to apply the multiple word tag for all the log text.
 - (OR)
 - i. Select **Reverse Single Word**.
 - j. Enter 1 in **Reverse Word Index** to mark the last word of the log text.
 - k. Similarly, you can enter any number to mark the word based on reverse order.
 - l. Click **Apply** to apply the reverse single word for all the log text.

Word Position Based Approach

Marking Type

☒ Single Word ☐ Multiple Words

Select	Start Word	Label
☒		date
☒		logtype

Word Position Based Approach

Marking Type

☐ Single Word ☒ Multiple Words

Select	Start Word	End Word	Label
☒			date
☒			logtype

Word Position Based Approach

Marking Type

☐ Single Word ☐ Multiple Words

☐ Reverse Single Word

Select	Reverse Word Index	Label
☒	1	date

Mark one apply all

Marking Type

☐ Index based ☐ Word Position based ☒ Prefix/Suffix based

Select	Start Tag	End Tag	Label
☒			ip_address
☒			log_date

13. If you select **Prefix/Suffix Based**, the Start Tag and End Tag fields appear.
 - a. Enter a word in **Start Tag** and another word in **End Tag**.
 - b. The words between start tag and end tag are marked in the log text.
 - c. Click **Apply** to apply the tags for all the log text.
14. The same markings will be applied for all the log text lines.
15. As a result, a green tick mark appears for all the lines.
16. Click **Done**.

SR#	Log Text	
1	2012-02-03 18:35:34 SampleClass6 [INFO] everything normal for id 577725851	✓
2	2012-02-03 18:35:34 SampleClass3 [DEBUG] detail for id 1304807656	✓
3	2012-02-03 18:35:34 SampleClass3 [WARN] missing id 423340895	✓
4	2012-02-03 18:35:34 SampleClass5 [TRACE] verbose detail for id 2082654978	✓
5	2012-02-03 18:35:34 SampleClass0 [ERROR] incorrect id 1886438513	✓
6	2012-02-03 18:35:34 SampleClass9 [TRACE] verbose detail for id 438634209	✓
7	2012-02-03 18:35:34 SampleClass8 [DEBUG] detail for id 2074121310	✓
8	2012-02-03 18:35:34 SampleClass0 [TRACE] verbose detail for id 1505582508	✓
9	2012-02-03 18:35:34 SampleClass0 [TRACE] verbose detail for id 1903854437	✓
10	2012-02-03 18:35:34 SampleClass7 [DEBUG] detail for id 915853141	✓

17. As a result, the regex format will be processed and a task will be created.

Run Task, Edit Task and Delete Task

To run, edit and delete task, follow the steps below:

1. In the left panel, click **Training Tasks**.
2. After you create regex, click **Run** to execute a task. For details, refer [Create Regex Task](#).
3. The status indicates the current status of the task. For Example: In Progress.
4. If required, click the **Edit** to edit the task again.
5. If required, click the **Delete** to delete the task from the task list.

The screenshot shows the ELIXIR Training Tasks interface. On the left is a sidebar with navigation links: Dashboard, Training Tasks (selected), Data Tasks, Configuration, and Help. The main area displays a table of training tasks. The table has columns: S.No., Task Name, Description, Result Type, Status, and Action. There are two tasks listed, both with a status of 'Completed'. The first task is 'DL_Approach' with result type 'ML Model'. The second task is 'task202152513324' with result type 'Regular Expression'. Each task has a set of action buttons: Run, Edit, Summary, Result, and Delete.

S.No.	Task Name	Description	Result Type	Status	Action
1	DL_Approach	DL_Approach	ML Model	Completed	Run Edit Summary Result Delete
2	task202152513324	task202152513324	Regular Expression	Completed	Run Edit Summary Result Delete

View Summary and Result

To view the execution summary and result, follow the steps below:

1. In the left panel, click **Training Tasks**.
2. After you run a task, click **Summary** to view the progress of execution.
3. After the execution gets completed, click **Result** to see the result of execution.

This screenshot is identical to the one above, showing the ELIXIR Training Tasks page with the same table of tasks.

The screenshot shows the 'Execution Summary' dialog box. It contains a list of progress bars for different tasks, showing the percentage of log_level completed and the estimated time to completion (ETA). The tasks are: loginfo, logdate, and task202152513324.

Task Name	Progress	ETA
loginfo	0.00% of log_level 0/32	ETA: 2 h, 38 m, 55 s
logdate	10.10% of log_level 2/32	ETA: 0 h, 1 m, 58 s
task202152513324	20.10% of log_level 6/32	ETA: 0 h, 0 m, 51 s
	32.50% of log_level 10/32	ETA: 0 h, 0 m, 43 s
	42.40% of log_level 13/32	ETA: 0 h, 0 m, 36 s
	50.30% of log_level 15/32	ETA: 0 h, 0 m, 33 s
	60.10% of log_level 16/32	ETA: 0 h, 0 m, 38 s
	70.10% of log_level 20/32	ETA: 0 h, 0 m, 38 s
	80.10% of log_level 24/32	ETA: 0 h, 0 m, 25 s
	90.10% of log_level 28/32	ETA: 0 h, 0 m, 15 s

The screenshot shows the 'Execution Result' dialog box. It contains a table with the results for the loginfo and logdate tasks, showing the TagName, Java Expression, and JS Expression.

TagName	Java Expression	JS Expression
loginfo	<code>\w++:[^@]*+</code>	<code>(?=(w+))1:(?=[^@*])2</code>
logdate	<code>[^:]*+:[^:]*+</code>	<code>(?=[^:]*+):1:(?=[^:]*+):2</code>

Data Tasks - Validation

To validate a log file, follow the steps below:

1. In the left panel, click **Data Tasks**.
2. In the **Data Tasks** page,
 - a. Select the **Log**.
 - b. Enter the **Log Separator**.
 - c. Click **Select Task**.
 - d. The List of Completed Tasks are displayed.
 - e. Select any of the completed tasks.
 - f. Click **Submit**.
 - g. As a result, the task is added in the **Data Tasks** page.
 - h. Click **Play** icon to validate the log file based on regex expression.
 - i. Click **View** icon to view the results.
 - j. If required, click the **Download** icon to download the csv file.
 - k. If required, click **Delete** icon to delete the validation.

ELIXIR
Accelerating Real Data Goals

Search

Dashboard

Training Tasks

Data Tasks

Configuration

Help

Fri, May 28 2021 - 06:51 am Version 0.9

Home / Data Task

Data Tasks

Log

--Select--

Log Separator

\n

Select Task

No file selected

Validation Name

validation_task_202152865120

Submit

S.No.	Date	Validation Name	Task Name	Approach	Status	Action
1	27/05/2021 07:13:13	validation_task_202152771218	task202152513324	Regular Expression	Completed	

API Reference

GET - Get Validation Task List

http://127.0.0.1:8000/api/list_validation_task

Response

```
{
  "Status": "Success",
  "Message": "Record fetched successfully",
  "Result": "[{\"validation_task_name\": \"Regex_Validation\", \"status\": \"Completed\", \"training_name\": \"task2021427102832\", \"training_type\": \"Regular Expression\"}]"
}
```

GET - Get Validation Task Result

http://127.0.0.1:8000/api/get_validation_result?task_id=23

Response

```
{
  "Status": "Success",
  "Message": "Record fetched successfully",
  "Result": "[
    {
      \"Log_No\": \"1\",
      \"ip_address\": \"192.168.4.164\",
      \"log_date\": \"07/19/2017 10:13:03\",
      \"log_level\": \"INFO\",
      \"log_number\": \"56441\"
    }
  ]"
}
```

Quick Start

Sample Log Content

192.168.4.164	07/19/2017	10:13:03	56441	INFO	executor.CoarseGrainedExecutorBackend: Registered signal handlers for [TERM, HUP, INT]
83.149.9.216	07/20/2017	10:13:07	56442	WARN	spark.SecurityManager: Changing view acls to: yarn,curi
24.236.252.67	08/21/2017	10:13:11	56443	ERROR	Changing modify acls to: yarn,curi
93.114.45.13	07/22/2017	10:13:15	56444	ALERT	SecurityManager: authentication disabled; ui acls disabled; users with view permissions: Set(yarn, curi); users with modify permissions: Set(yarn, curi)
66.249.73.135	07/23/2017	10:13:19	56445	DEBUG	Changing view acls to: yarn,curi
50.16.19.13	07/24/2017	10:13:23	56446	INFO	executor.CoarseGrainedExecutorBackend: Connecting to driver: spark://CoarseGrainedScheduler@10.10.34.11:48069
66.249.73.185	07/25/2017	10:13:27	56447	WARN	executor.CoarseGrainedExecutorBackend: Got assigned task 0
46.105.14.53	07/26/2017	10:13:31	56448	ERROR	Remoting: Starting remoting
110.136.166.128	08/01/2017	10:13:35	56449	ALERT	Remoting: Remoting started; listening on addresses :[akka.tcp://sparkExecutorActorSystem@mesos-slave-07:55904]
123.125.71.35	08/02/2017	10:13:39	56450	DEBUG	storage.MemoryStore: MemoryStore started with capacity 17.7 GB
50.150.204.184	08/03/2017	10:13:43	56451	INFO	executor.Executor: Running task 0.0 in stage 0.0 (TID 0)
207.241.237.225	08/04/2017	10:13:47	56452	WARN	storage.MemoryStore: Block broadcast_9_piece0 stored as bytes in memory (estimated size 5.2 KB, free 5.2 KB)
200.49.190.101	08/05/2017	10:13:51	56453	ERROR	spark.CacheManager: Partition rdd_2_1 not found, computing it
200.49.190.100	08/06/2017	10:13:55	56454	ALERT	rdd.HadoopRDD: Input split: hdfs://10.10.34.11:9000/pjhe/logs/2kSOSP.log:21876+7292
66.249.73.185	08/07/2017	10:13:59	56455	DEBUG	broadcast.TorrentBroadcast: Started reading broadcast variable 8
66.249.73.135	08/08/2017	10:14:03	56456	INFO	broadcast.TorrentBroadcast: Reading broadcast variable 8 took 19 ms
67.214.178.190	08/09/2017	10:13:07	56457	WARN	Configuration.deprecation: mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
207.241.237.220	08/10/2017	10:13:11	56458	ERROR	storage.MemoryStore: Block broadcast_4_piece0 stored as bytes in memory (estimated size 95.0 B, free 317.6 KB)
46.105.14.53	08/11/2017	10:13:15	56459	ALERT	python.PythonRunner: Times: total = 1072, boot = 856, init = 210, finish = 6
207.241.237.227	08/12/2017	10:13:19	56460	DEBUG	storage.MemoryStore: Block rdd_2_2 stored as bytes in memory (estimated size 850.0 B, free 320.3 KB)

91.177.205.119 08/13/2017 10:13:03 66441 INFO spark.CacheManager: Partition rdd_11_2 not found, computing it
66.249.73.185 08/24/2017 10:13:07 66442 WARN 17/06/09 20:10:53 INFO storage.MemoryStore: Block broadcast_16 stored as values in memory (estimated size 9.8 KB, free 432.0 KB)
207.241.237.228 08/15/2017 10:13:11 66443 ERROR broadcast.TorrentBroadcast: Reading broadcast variable 16 took 13 ms
207.241.237.101 08/16/2017 10:13:15 66444 ALERT storage.MemoryStore: Block broadcast_16_piece0 stored as bytes in memory (estimated size 5.6 KB, free 422.1 KB)
87.169.99.232 08/17/2017 10:13:19 66445 DEBUG executor.Executor: Running task 4.0 in stage 7.0 (TID 319)
209.85.238.199 08/18/2017 10:13:23 66446 INFO executor.CoarseGrainedExecutorBackend: Got assigned task 319
81.220.24.207 09/01/2017 10:13:27 66447 WARN broadcast.TorrentBroadcast: Started reading broadcast variable 16
46.105.14.53 09/03/2017 10:13:31 66448 ERROR executor.Executor: Running task 3.0 in stage 7.0 (TID 318)
66.249.73.135 09/05/2017 10:13:35 66449 ALERT executor.CoarseGrainedExecutorBackend: Got assigned task 312
218.30.103.62 09/07/2017 10:13:39 66450 DEBUG executor.Executor: Running task 1.0 in stage 7.0 (TID 299)
74.125.40.20 09/09/2017 10:13:43 76451 INFO executor.CoarseGrainedExecutorBackend: Got assigned task 299
71.212.224.97 09/11/2017 10:13:47 76452 WARN executor.Executor: Running task 0.0 in stage 7.0 (TID 285)
218.30.103.62 09/13/2017 10:13:51 76453 ERROR executor.CoarseGrainedExecutorBackend: Got assigned task 285
108.174.55.234 09/15/2017 10:13:55 76454 ALERT executor.Executor: Finished task 1.0 in stage 6.0 (TID 261). 2087 bytes result sent to driver
121.107.188.202 09/17/2017 10:13:59 76455 DEBUG python.PythonRunner: Times: total = 42, boot = -101, init = 143, finish = 0
218.30.103.62 09/19/2017 10:14:03 76456 INFO storage.BlockManager: Found block rdd_2_1 locally
198.46.149.143 09/21/2017 10:13:07 76457 WARN storage.MemoryStore: Block broadcast_15 stored as values in memory (estimated size 9.2 KB, free 416.6 KB)
66.249.73.135 09/25/2017 10:13:15 76459 ALERT broadcast.TorrentBroadcast: Reading broadcast variable 15 took 15 ms
86.1.76.62 10/05/2017 10:13:19 76460 DEBUG executor.Executor: Running task 4.0 in stage 6.0 (TID 279)
66.249.73.135 10/10/2017 10:13:39 76450 DEBUG executor.Executor: Finished task 1.0 in stage 5.0 (TID 220). 2272 bytes result sent to driver
107.170.41.69 10/15/2017 10:13:43 86451 INFO python.PythonRunner: Times: total = 42, boot = -79, init = 120, finish = 1

50.16.19.13	10/20/2017	10:13:47	86452	WARN	storage.BlockManager: Found block rdd_2_0 locally
46.105.14.53	11/07/2017	10:13:51	86453	ERROR	broadcast.TorrentBroadcast: Reading broadcast variable 14 took 12 ms
208.115.111.72	11/18/2017	10:13:55	86454	ALERT	storage.BlockManager: Found block rdd_2_4 locally
50.180.79.170	11/27/2017	10:13:59	86455	DEBUG	storage.MemoryStore: Block broadcast_14_piece0 stored as bytes in memory (estimated size 5.8 KB, free 392.3 KB)
208.115.111.72	12/12/2017	10:14:03	86456	INFO	executor.CoarseGrainedExecutorBackend: Got assigned task 206
105.235.130.196	12/25/2017	10:13:07	86457	WARN	python.PythonRunner: Times: total = 42, boot = -125, init = 166, finish = 1
174.37.205.76	12/29/2017	10:13:11	86458	ERROR	python.PythonRunner: Times: total = 39, boot = -123, init = 162, finish = 0
54.255.13.204	01/04/2018	10:13:15	86459	ALERT	python.PythonRunner: Times: total = 41, boot = 16, init = 25, finish = 0
54.255.13.204	01/11/2018	10:13:19	86460	DEBUG	storage.MemoryStore: Block rdd_6_4 stored as bytes in memory (estimated size 287.0 B, free 369.5 KB)

Sample 50 lines provided for reference..

How to upload Log File and Generate Regular Expression?

Training Task

1. In **Tasks**, create a task by selecting **Regular Expression** in the **Type** field.
2. Click **Import** and select the log file.
3. As a result, the log text will be listed in the Create Task page.
4. Click any one + icon on the right side of log text.
5. The **Mark Tags** window appears.
6. To create a tag, click + icon on the right side.
7. Enter the name for the tag and click **Update**. The tag gets added below the Tags section.
8. Select any portion of the log text, for example: log_date.
9. Click + icon on the right side of the tag created to mark the selected portion.
10. Click **Apply All** to mark the same for all the log texts listed.
11. Now, the task gets added in the **Task List**.
12. Click **Run** to execute the task.
13. Click **Summary** to view the Execution Summary for both Completed and In-Progress tasks.
14. Click **Result** to view the result in the form of Regular Expression for Completed task.
15. As a result, the dashboard page gets updated.

Data Task

In **Data Tasks** page, follow the steps below to validate the completed tasks.

1. Select the log file and select any completed task.
2. Enter the validation name and click **Submit**.
3. As a result, a validation task gets added.
4. Click **Run** icon to run the validation.
5. Click **View** icon to view the result.
6. Click **Download** icon to download the csv file.